This article has been accepted for publication in IEEE Journal of Emerging and Selected Topics in Power Electronics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JESTPE.2022.3207454

Data-Driven Finite Control-Set Model Predictive Control for Modular Multilevel Converter

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***Abstract*—This paper investigates a data-driven based predictive current control (DD-PCC) approach for a modular multilevel converter (MMC) to circumvent the sensitiveness to parameter variation and unmodeled dynamics of finite control-set model predictive control (FCS-MPC) method. By integrating a model-free adaptive control based data-driven solution into the FCS-MPC framework, the performance deterioration caused by model uncertainties is suppressed. The design of the suggested controller is only based on input-output measurement data, where neither the parameter information nor the knowledge of detailed MMC models are required, leading to improved robustness against parameter drifts and model uncertainness. Moreover, a simplified cost function formula that takes into account output current tracking and circulating current regulation is constructed to efficiently determine the optimal insertion index of each arm. Finally, simulation and experimental results are obtained to verify the steady-state, dynamics, and robustness performance of the proposed approach.**

***Index Terms*—Finite control-set model predictive control (FCS-MPC), data-driven control, modular multilevel converter (MMC), robustness.**

I. Introduction

Model predictive control (MPC) method gains increasing attention in industrial applications due to its distinctive features of straightforward integration of system non-linearities and operating constraints. Meanwhile, powerful modern microprocessors and the latest achievements in terms of control have expanded the application of the MPC method to machine drives and complex multilevel power converter systems like modular multilevel converter (MMC), among others [1]–[3].

Basically, the MPC method can be categorized into two subbranches, i.e., continuous control set MPC (CCS-MPC) and finite control set MPC (FCS-MPC). In the former, the continuous control inputs are obtained via minimization of the cost function representing the tracking error of the controlled variables. The desired switching states are synthesized through a modulator, resulting in a fixed switching frequency [4]. As an alternative, the FCS-MPC technique takes into account the discrete nature of the power converter to formulate the MPC algorithm. The desired switching states are directly selected by enumerating all the candidate options while the modulation process is eliminated. The comparative studies between different MPC methods have been investigated in [5]–[7]. As evidenced in the literature, the CCS-MPC can address long prediction horizon problems, whereas the FCS-MPC typically considers a prediction horizon of one step. Nevertheless, owing to its prominent features such as direct manipulation of converter switching combinations, the capability of handling multiple constraints, and the possibility to combine different control variables simultaneously, the FCS-MPC approach remains the predominant solution in power converters systems with multiple control variables, e.g., MMC.

An early work in FCS-MPC regulated MMC was presented in [8] where the framework of the FCS-MPC approach for a three-phase MMC has been established. To circumvent the challenge of the explosive increased computational burden of the FCS-MPC method with higher voltage levels, various modified control methods were introduced to ease its implementation [9]–[11]. Besides, different control architectures were presented to cope with the multiobjective optimization problem such as two-stage method [12], variable rounding level control [13], or sequential MPC method [14]. Despite the significant achievements of existing methods, as a model-based control method, the control performances of the FCS-MPC method are greatly dependent on the knowledge of used parameters and accurateness of the constructed models [15], [16]. In reality, however, the unmodeled dynamics are inevitable, and parameter values in the physical systems may be affected by factors, e.g. operating point, temperature, and external disturbances [17], [18]. Accordingly, FCS-MPC faces the inherent challenges of model sensitiveness and parameter variation in real-world applications, especially due to the complex structure and the large number of parameters [19].

Various solutions have been suggested in the literature for the problem of parameter sensitiveness of the FCS-MPC approach. Distinctive online parameter identification methods were proposed to compensate for the mismatch between actual parameters and those used in the controller. In [20], a sensorless method with robust adaptive inductance estimation was proposed. In this paper, a parameter estimation technique in terms of model reference adaptive system was deployed to

This work was supported in part by the National Natural Science Foundation of China under Grant 51807177, 51827810, and in part by China Postdoctoral Science Foundation under Grant 2020M681855, and in part by Natural Science Foundation of Zhejiang Province under Grant LY21E070004 and Grant LY22E070003. J. Rodriguez acknowledges the support of ANID through projects FB0008, 1210208 and 1221293. (*Corresponding authors: Lin Qiu.*)

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identify the value of grid-side inductance. However, the online parameter identification procedure leads to a large number of calculations, which aggravates the computational burden of the FCS-MPC controller.

Another effective way to enhance the robustness performance against the variations of the parameters is to develop a model-free design [21]–[24]. In contrast to the model-based design method, the knowledge of the plant model or the detailed parametric information is not required in the model-free design framework, which relaxes the dependency on unknown model parameters of the FCS-MPC approach. The authors in [21] is dedicated to develop a parameter-free FCS-MPC by deploying a recursive least-square self-commissioning model for synchronous motor drives. In [22], an extended state observer (ESO) was embedded into the FCS-MPC framework to estimate the modeled and unmodeled part of the permanent magnet synchronous motor, achieving enhanced robustness while not involving parameter information. A predictor-based neural network (PNN) FCS-MPC method was developed in [23] to regulate the active front-end MMC, where neural adaptation laws were incorporated to realize the fast identification of system dynamics. Similarly, the model uncertainties were estimated in [24] by deploying a fuzzy approximation algorithm. Due to the fact that no parameter values are required in these designs, improved robustness and reliability can be achieved in the presence of parametric uncertainties. However, the dependence on hypothetical informations from the initial controller can be considered as a weak point of these strategies.

Aiming to further nullify the effect of inaccurate models in the FCS-MPC method, data-driven control emerges as an alternative solution to formulate the controller. A data-driven recursive least squares model identification algorithm was utilized in motor drive applications to improve the prediction accuracy [25]. Alteratively, a data-driven neural predictors-based robust MPC was proposed for MMC by using the real-time and historical data [26]. This trend has also been witnessed in the case of CCS-MPC [27]. The emerging works revel that this research topic remains an ongoing process.

The aforementioned observations motivate us to explore a fundamentally different architecture to cope with the parametric uncertainties and system nonlinearities in the FCS-MPC approach. More specifically, a model-free adaptive control (MFAC) based data-driven concept is embedded into the FCS-MPC framework to solve the inherent problem of parameter drift and model inaccuracy, creating a new architecture named data-driven-based predictive current control (DD-PCC). In contrast to conventional FCS-MPC methods, the proposed data-driven approach allows for directly manipulating the control actions without using the accurate description of the models and the knowledge of physical parameters. Accordingly, the adaptability of the FCS-MPC controller to different parameter conditions can be guaranteed.

The major novelty of this work results from the following contributions:

1) This paper established a data-driven predictive control architecture by integrating the model-free adaptive control concept into the FCS-MPC framework. The implementation of the suggested solution is only based on the I/O measurement data of the FCS-MPC controller without using the expert knowledge of the explicit model and system parameters.
2) This paper designed a simplified cost function formulation to facilitate the easy implementation of the proposed method.
3) To verify the effectiveness of this proposal, a multiobjective case study of MMC is provided. The improved control performance is confirmed by simulation and experimental results.

For the purpose of clear methodology, this paper is organized as follows: the conventional FCS-MPC approach of MMC is briefly presented in Section II; the data-driven based predictive current control is proposed in Section III; Section IV and Section V provide the simulation and experimental results on an MMC-based inverter to confirm its competency in robustness and reliability; and finally, Section VI concludes this paper.

II. The existing FCS-MPC approach for MMC

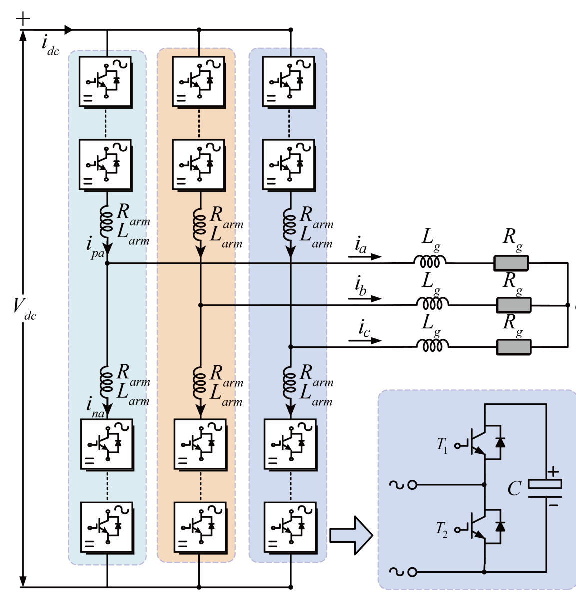

Fig. 1. Common structure of a three-phase MMC.

The common topology of a three-phase MMC is shown in Fig. 1. According to the circuit diagram, the voltage dynamic phase-x, $x \in \{a, b, c\}$, is described as [28]

$$\frac{U_{dc}}{2} = u_{px} + R_{arm}i_{px} + L_{arm}\frac{di_{px}}{dt} + R_g i_x + L_g\frac{di_x}{dt} \quad (1)$$

$$-\frac{U_{dc}}{2} = -u_{nx} - R_{arm}i_{nx} - L_{arm}\frac{di_{nx}}{dt} + R_g i_x + L_g\frac{di_x}{dt} \quad (2)$$

$$i_x = i_{px} - i_{nx} \quad (3)$$

where U_{dc} and i_x represent the dc-link voltage and ac-side output current of phase-x, respectively. L_g and R_g denote the ac-side inductor and resistor, respectively. L_{arm} and R_{arm} denote the arm inductor and resistor, respectively. u_{px}/u_{nx}

and i_{px}/i_{nx} denote the upper/lower arm voltage and current, respectively.

Based on (1)-(3), the dynamics of phase-x ac-side output current and circulating current can be deduced as [29]

$$\begin{cases} \dfrac{di_x}{dt} = -\dfrac{R_z}{L_z} i_x + \dfrac{1}{2L_z} v_\Delta \\ \dfrac{di_{zx}}{dt} = -\dfrac{R_{arm}}{L_{arm}} i_{zx} + \dfrac{V_{dc} - v_\Sigma}{2L_{arm}} \end{cases} \tag{4}$$

where i_{zx} denotes the circulating current of phase-x, $L_z = L_g + \frac{1}{2}L_{arm}$ and $R_z = R_g + \frac{1}{2}R_{arm}$ are equivalent inductance and resistance, respectively. v_Δ and v_Σ are the control input of output current and circulating current of phase-x, denoted as $v_\Delta = u_{nx} - u_{px}$, $v_\Sigma = u_{nx} + u_{px}$, respectively.

The discrete-time model is deduced by considering the first-order Euler discretization for the model in (4), as given by

$$\begin{cases} i_x^p(k+1) = (1 - \dfrac{R_z T_s}{L_z}) i_x(k) + \dfrac{T_s}{2L_z} v_\Delta(k) \\ i_{zx}^p(k+1) = (1 - \dfrac{R_{arm} T_s}{L_{arm}}) i_{zx}(k) + \dfrac{V_{dc} - v_\Sigma(k)}{2L_{arm}} T_s \end{cases} \tag{5}$$

where $i_x^p(k+1)$ and $i_{zx}^p(k+1)$ denote the predicted value of the output current and circulating current, respectively.

For the commonly used voltage level-based FCS-MPC regulated MMC, the capacitor voltages are manipulated by an independent balancing method. The output currents and circulating current are regulated by the following cost function:

$$J = |i_x^*(k+1) - i_x^p(k+1)| + \theta_z |i_{zx}^*(k+1) - i_{zx}^p(k+1)| \tag{6}$$

where $|\cdot|$ denotes the absolute value of a scalar quantity, superscript "$*$" denotes the reference value, θ_z is weighting factor.

It is worth noting that the performance of the conventional model-based FCS-MPC method largely depends on the precise prediction of future behaviors via the discrete-time model (5). However, due to the fact that unmodeled dynamics and parameters variation are not taken into consideration in (5), in the presence of inaccuracy parameter values and unmodeled disturbance, the performance deterioration becomes inevitable. To circumvent this drawback, a novel data-driven based model-free FCS-MPC methodology is developed in the following section.

III. Data-driven based predictive current control

A. Data-Driven Model-Free Adaptive Controller

The controller of the model-based control method is designed based on the mathematical model of a real plant. Conversely, the data-driven control, as a model-free control method, is directly implemented from the input-output data of the regulated system or knowledge from data processing. As a result, data-driven control is capable of overcoming the model dependence issues of the conventional model-based control method. This prominent feature endorses the data-driven approach to be an alternative solution for the system control problem, particularly for dealing with system uncertainties [30].

Among various data-driven methods, the model-free adaptive control (MFAC) scheme shows good potential to cope with the complex control problem. The prominent features of the MFAC method include low-cost, easy implementation, and strong robustness. These distinctive merits motivate us to explore an MFAC-based data-driven predictive current control method to fulfill the control demand of an MMC, as elaborated below.

The discrete-time dynamics in (5) can be rewritten as

$$y(k+1) = f\left(y(k), \cdots, y(k-n_y), u(k), \cdots, u(k-n_u)\right) \tag{7}$$

where $y(k+1) \in R$ and $u(k) \in R$ denote the system output and control input in terms of i_x and v_x in this design, respectively, and $f(\cdots) :^{n_u+n_y+2} \rightarrow R$ is an unknown nonlinear function.

Under two assumptions in the literature [30], there must exists a time-varying parameter $\phi(k) \in R$, called pseudo partial derivative, such that the nonlinear system model (7) can be transformed into the dynamic linearization (DL) data model including compact form DL (CFDL), partial-form DL (PFDL) and the full-form DL (FFDL) data model. From the computational complexity point of view, the CFDL data model is used in this paper [31]:

$$\Delta y(k+1) = \phi(k) \Delta u(k) \tag{8}$$

where $\Delta y(k+1) = y(k+1) - y(k)$ and $\Delta u(k) = u(k) - u(k-1)$ are the change of output variable and the change of control input between two consecutive time instants, respectively.

Consider the following control input criterion function:

$$J(u(k)) = |y^*(k+1) - y(k+1)|^2 + \lambda |\Delta u(k)|^2 \tag{9}$$

where λ is a weighting factor.

Substituting CFDL data model (8) into cost function (9) and differentiating (9) with respect to $u(k)$ and setting it to be zero yields

$$u(k) = u(k-1) + \frac{\rho \phi(k)}{\lambda + |\phi(k)|^2} \left(y^*(k+1) - y(k)\right) \tag{10}$$

where $\rho > 0$ is a step factor.

Similarly, the index function for the unknown pseudo-partial derivative estimation is defined as

$$\begin{aligned} J(\hat{\phi}(k)) = &\left| y(k) - y(k-1) - \hat{\phi}(k) \Delta u(k-1) \right|^2 \\ &+ \mu |\hat{\phi}(k) - \hat{\phi}(k-1)|^2 \end{aligned} \tag{11}$$

where μ is a weighting constant.

Using optimal condition $\partial J / \partial \hat{\phi}(k) = 0$, the time-varying parameter $\phi(k)$ can be estimated by

$$\begin{aligned} \hat{\phi}(k) = &\hat{\phi}(k-1) + \frac{\eta \Delta u(k-1)}{\mu + |\Delta u(k-1)|^2} \\ &\times \left(\Delta y(k) - \hat{\phi}(k-1) \Delta u(k-1) \right) \end{aligned} \tag{12}$$

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where

$$\hat{\phi}(k) = \hat{\phi}(1)\ if \left|\hat{\phi}(k)\right| \le \varepsilon\ or\ |\Delta u(k-1)| \le \varepsilon \quad or\ \mathrm{sign}(\hat{\phi}(k)) \ne \mathrm{sign}(\hat{\phi}(1)). \tag{13}$$

Remark: The selection of the control parameters in this design is elaborated as follows. λ and μ are two positive numbers used to limit the change of control input and pseudo-partial derivative, respectively. Particularly, a smaller λ leads to faster system response but may produce overshoot, and vice versa. The step factor $\eta \in (0, 2]$ and $\rho > 0$, are added to make the controller algorithm more flexible. The reset mechanism in (13) is to endow parameter estimation algorithm with a strong ability to track the time-varying parameter. $\hat{\phi}(1)$ is the initial value of $\hat{\phi}(k)$, ε denotes a small positive constant selected as $1e^{-5}$ in this design. The detailed stability analysis can refer to [31].

B. Proposed DD-PCC Controller

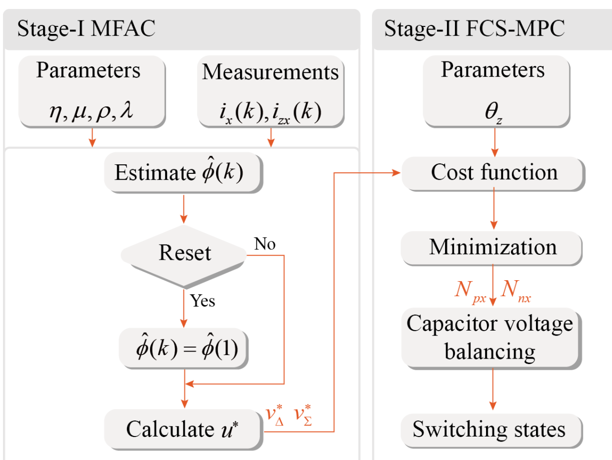

Fig. 2. Flowchart of the proposed DD-PCC scheme.

The flowchart of the proposed DD-PCC scheme is shown in Fig. 2. In the proposed approach, the implementation of the DD-PCC method mainly consists of two stages, namely Stage-I and Stage-II. In Stage-I, the MFAC solution is deployed to determine the optimal control input only based on the input-output measurement data of the FCS-MPC controller in the absence of necessary expert knowledge of the explicit model and system parameters. In Stage-II, the multiple control objectives in terms of output current tracking and circulating current elimination are considered in the FCS-MPC framework. The optimum insertion indexes of upper and lower arm are obtained by enumerating all candidate combinations. Finally, the desired switching state is selected by employing the sorting-based capacitor voltage balancing method.

The implementation of the proposed DD-PCC method involves several major steps, as explained below.

1) Measure the output current i_x and circulating current i_{zx} at the k-th sampling instant.
2) Estimate the pseudo partial derivative $\hat{\phi}(k)$ by using (12). It is noteworthy that if the reset conditions in (13) are satisfied, the estimated $\hat{\phi}(k)$ should be reset to $\hat{\phi}(1)$.

TABLE I
SYSTEM PARAMETERS

Variable	Description	Experimental	Simulation
R_g	Output resistance (Ω)	0.1	0.5
L_g	Output inductance (mH)	10	5
R_{arm}	Arm resistance (Ω)	0.2	0.2
L_{arm}	Arm inductance (mH)	2.5	10
V_{dc}	DC bus voltage (V)	500	3300
C	SM capacitance (μF)	1600	6600
N	SMs per arm	4	10
T_s	Sampling period (μs)	100	100

3) Calculate the optimal control inputs of the MMC based on (10).
4) Formulate simplified cost function J including the optimal control inputs and the candidate voltage vectors as given by

$$J = |v_\Delta^j - v_\Delta^*| + \theta_z |v_\Sigma^j - v_\Sigma^*| \tag{14}$$

where v_Δ^* and v_Σ^* denote reference voltage vectors of output current and circulating current obtained from data-driven controller. v_Δ^j and v_Σ^j represent the candidate voltage combinations in each arm.
5) Select the desired insertion index of each arm by enumerating all candidate voltage combinations in each arm, as described by

$$\begin{cases} v_\Delta^j = \dfrac{U_{dc}}{N} N_g \\ v_\Sigma^j = N_{px}\overline{v}_{px} + N_{nx}\overline{v}_{nx} \end{cases} \tag{15}$$

where N_{px}, N_{nx} and $\overline{v}_{px}$, $\overline{v}_{nx}$ denote the insertion index and average voltage of upper and lower arm, respectively. N_g represents the presentable output voltage-level, as given by

$$N_g = \{-N, -(N-2), ..., 0, ..., (N-2), N\} \tag{16}$$

where N denotes the number of submodules in each arm.
6) Employ the optimal switching state by using the sorting-based capacitor voltage balancing method in [10].

As depicted in (10), the controller design process of the proposed method only involves the input-output data of the system instead of the practical models and parameters, resulting in adaptability and robustness against model uncertainties and parameter variation. Nevertheless, it is worthwhile to point out that the induced adjustable parameters and additional computational cost are issues that have to be considered in this proposal, which will be taken into account in our further work.

IV. Simulation Result

To verify the steady-state, transient-state, and robustness performance of the proposed DD-PCC method, comparative simulation studies of the conventional MPC method and the proposed solution on a three-phase MMC are carried out under MATLAB/SIMULINK environment, and related parameters are listed in Table. I.

The steady-state performance of the conventional MPC scheme and the proposed DD-PCC solution is demonstrated in Fig. 3 and Fig. 4, respectively. As illustrated in the figures,

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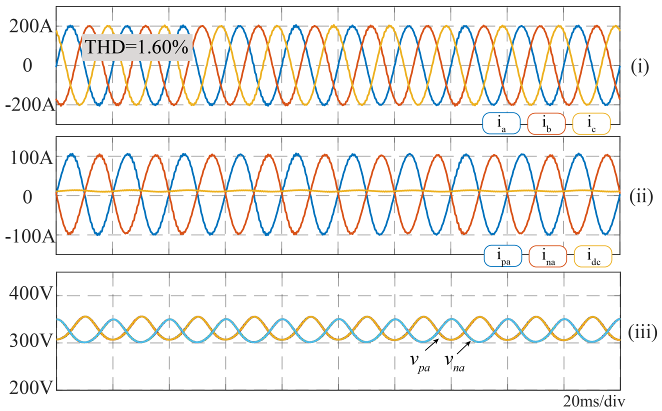

Fig. 3. Simulation results of the conventional MPC method: steady-state.

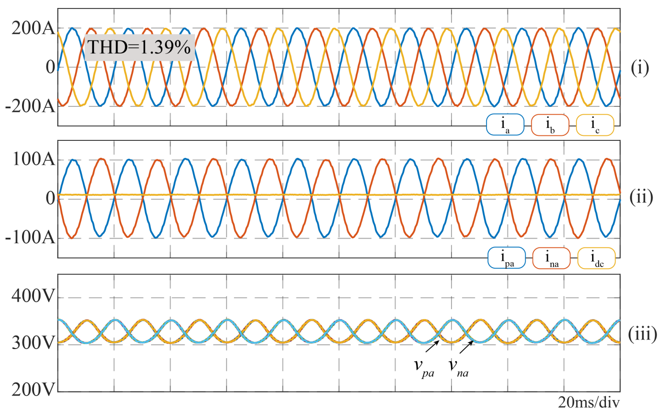

Fig. 4. Simulation results of the proposed DD-PCC solution: steady-state.

both the conventional MPC method and the proposed one are capable of generating sinusoidal waveforms of the three-phase output current, while the arm currents, dc-link current, and capacitor voltages are also well regulated. It is noteworthy that, with the identical operation condition, the proposed DD-PCC solution shows a comparable even better control performance in terms of lower total harmonic distortion (THD) of three-phase output current as compared with the conventional MPC method, which demonstrates good steady-state performance of the proposed DD-PCC method.

Fig. 5 describes the dynamic response of the proposed DD-PCC solution. As illustrated in Fig 5(a), with a step change in the reference current magnitude, the proposed DD-PCC solution follows the reference trajectory accurately. Alternatively, the transient performance of the proposed DD-PCC solution with a step change in the reference current frequency is recorded in Fig. 5(b). To further illustrate the transient performance of the DD-PCC solution, the dynamic response of the proposed solution and the conventional method is compared in detail in Fig. 6. As indicated in Fig. 6, both schemes achieve a very fast dynamic response when the reference current magnitude changes suddenly, whereas the tracking error of the proposed solution is slightly better than that of the conventional MPC method, as shown in Fig.

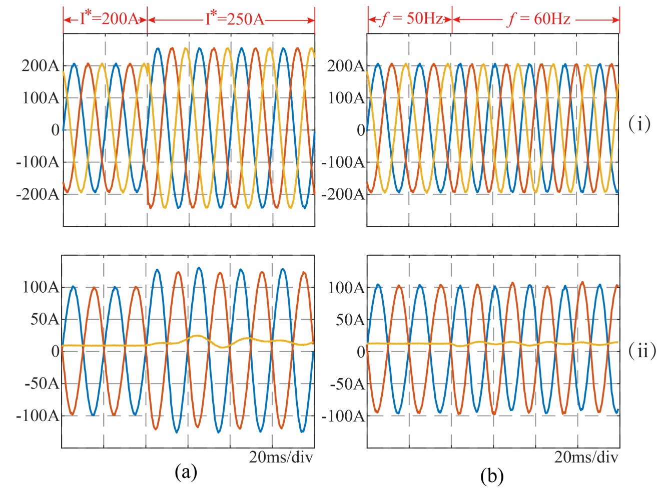

Fig. 5. Simulation results of the suggested DD-PPC: transient-state. (a) step change in reference current magnitude and (b) step change in reference current frequency [(i) output currents, (ii) arm currents and dc-link current].

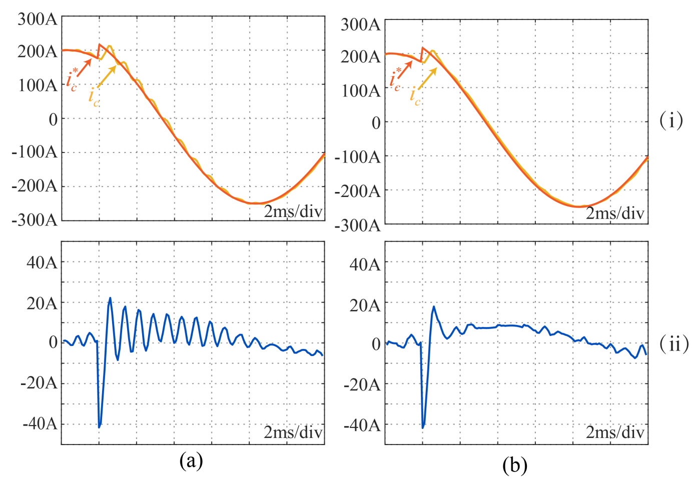

Fig. 6. Comparison results of dynamic response: (a) conventional MPC and (b) proposed DD-PCC [(i) reference current and output current, (ii) tracking error].

6(b)(ii).

To verify the robustness performance of the proposed DD-PCC method, a comparison simulation with the conventional MPC method under parameter mismatch condition is carried out, as demonstrated in Fig. 7. Initially, the conventional MPC method is deployed, the controlled variables are well regulated, The THD of the output current is 1.60%, as given in Table. II. However, when parameter mismatch occurs, the inductance is reduced by 40% (L_g = 5mH $\rightarrow$ 3mH) while the parameter used in the controller remains constant. Due to the model mismatch, the predicted future behavior of the controlled variables becomes inaccurate. Consequently, the performance of the conventional MPC is obviously degraded with a THD value of 11.55% and ripples appear on the output current. At the same time, the tracking error (e%) is increased to 9.36% as appreciated in Table. II. However, when the proposed DD-PCC solution is activated, the deteriorated

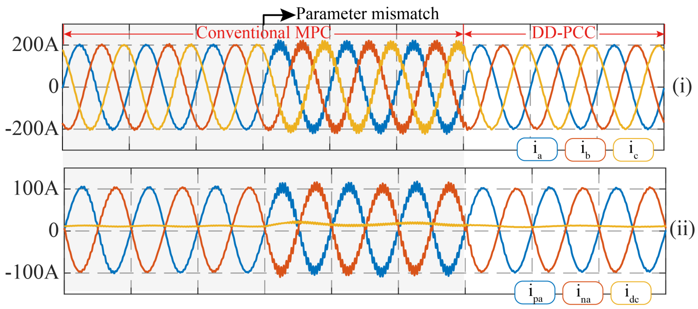

Fig. 7. Comparison results considering parameter mismatch. [(i) output currents, (ii) arm currents and dc-link current].

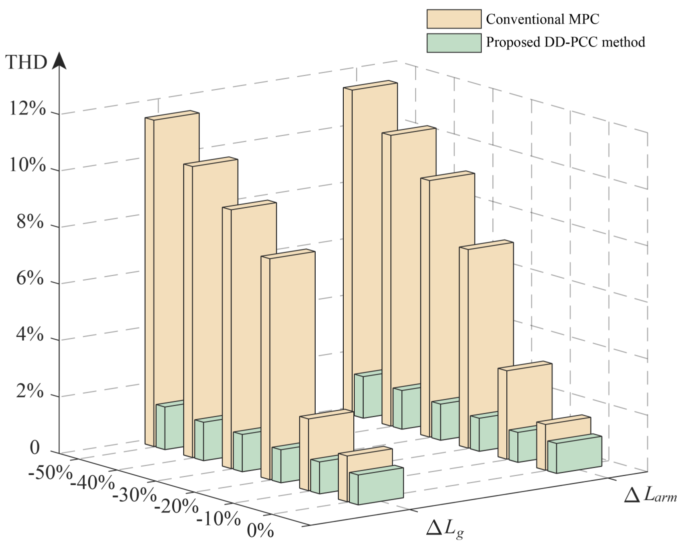

Fig. 8. Robust performance comparison of the conventional MPC method and the proposed DD-PCC solution.

control performance caused by parameter drift is greatly improved with a low current distortion, the THD is 1.56% and tracking error is 3.81% at this time, which shows good robustness performance of the proposed DD-PCC solution. The proposed method also yields a relative lower switching frequency, which stems from the restriction of the control input criteria in (9). A computation time (T_e) comparison between two MPC methods is also demonstrated in Table. II. It should be noted that the price that has to be paid for the improved robustness performance of the proposed DD-PCC solution is the additional estimation of pseudo-partial derivative, which increases the computation time of the proposed method. On the other hand, the simplified cost function formulated in this design yields a favorable influence on the calculation cost. In this regard, the computation time of the proposed method is slightly increased as compared with the conventional MPC method.

The robustness of the proposed DD-PCC solution can be further validated by Fig. 8, wherein the performance metric in terms of the THD of output current are evaluated with different filter inductance and arm inductance. Fig. 8 illustrates that the control performance of the conventional MPC method deteriorates significantly when system parameters are changed. However, the proposed DD-PCC solution shows better robustness against parameter variation. Consequently, the proposed DD-PCC solution inherits the advantages of the fast dynamic response of the traditional MPC method while exhibits strong robustness against parameter variations, which is a clear merit over the classical breed of MPC methods.

V. Experimental Verification

TABLE II
COMPARISON RESULT OF TWO METHODS

Control method	THD		$e\%$	T_e	SF
	$L_g = 5mH$	$L_g = 3mH$	$L_g = 3mH$		
Conventional method	1.60%	11.55%	9.36%	55.02μs	1261
Proposed DD-PCC	1.39%	1.56%	3.81%	55.36μs	816

In this section, comparative experiment studies are conducted on a prototype MMC inverter connected to a passive load to verify the effectiveness of the proposed DD-PCC solution. The specifications are summarized in Table I. For the implementation of this test, a TMS320C28346 digital signal processor (DSP) is deployed as the controller.

A. Steady-state Analysis

The steady-state performance of the suggested DD-PCC method is experimentally compared with the conventional MPC scheme in Fig. 9. Fig. 9(a) shows the experimental results of the conventional MPC scheme, while the control performance of the proposed DD-PCC solution is recorded in Fig. 9(b). As indicated in Fig. 9(i) and (ii), under identical operating conditions, both the conventional method and the proposed one can generate sinusoidal output currents, and arm currents are well regulated. The capacitor voltages of SMs are also regulated to their nominal values, as depicted in Fig. 9(iii). The comparison results show that the proposed DD-PCC approach can achieve comparable performance with the conventional FCS-MPC method, while only I/O data of the MMC is involved in the controller design process instead of precise model and parametric information.

B. Transient-state Analysis

The dynamic performance of the proposed DD-PCC is experimentally demonstrated in Fig. 10. Fig. 10(a) illustrates the transient-state performance of the proposed DD-PCC method with a step change in the amplitude of reference output current. The good tracking performance of output current is illustrated in Fig. 10(a)(i). The upper/lower arm currents are increased with the output current magnitude, as recorded in Fig. 10(a)(ii). Fig. 10(b) depicts the step response of the proposed controller, where the reference frequency of output current is increased from 50 Hz to 60 Hz. As given in Fig. 10(b)(i), fast and accurate tracking of the output current can be achieved when the frequency of output current is suddenly changed. Meanwhile, the tracking performance of arm currents are given in Fig. 10(b)(ii). As appreciated in Fig. 10, the proposed DD-PCC method achieves satisfactory control performance without using the detailed model and

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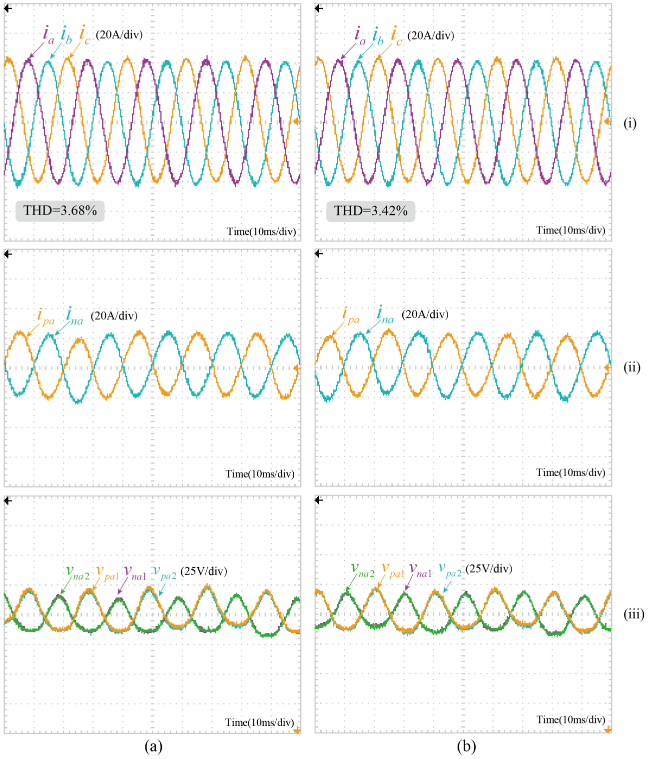

Fig. 9. Steady-state performance: (a) conventional MPC and (b) proposed DD-PCC. [(i) three-phase output currents, (ii) upper and lower arm current of phase-a, (iii) SMs voltage of phase-a.]

parameter values, whilst inheriting a high dynamic response from FCS-MPC.

C. Robustness Performance Analysis

In order to verify the robustness performance of the proposed DD-PCC controller, comparative experimental studies are conducted with the conventional MPC method and ESO-based method under mismatched parameter conditions. To simulate parameter drift, the grid-side inductance is selected as 8mH, while the parameter used in the conventional MPC controller remains 10mH. Accordingly, in this scenario, the predictive model of the conventional MPC method becomes inaccurate, leading to an obvious deterioration of the control performance, as recorded in Fig. 11(a). However, as shown in Fig. 11(b), the proposed solution and ESO-based MPC exhibit a comparable performance, which is slightly degraded under the same condition. The successful suppression of performance deterioration caused by parameter variations lies in the fact that both model information and parametric value are not involved in the proposed controller designing process.

VI. Conclusion

This paper investigated the possibility of combining a model-free adaptive control-based data-driven control concept with a finite-set model predictive current control method for the regulation of MMC. In the proposed design, the output

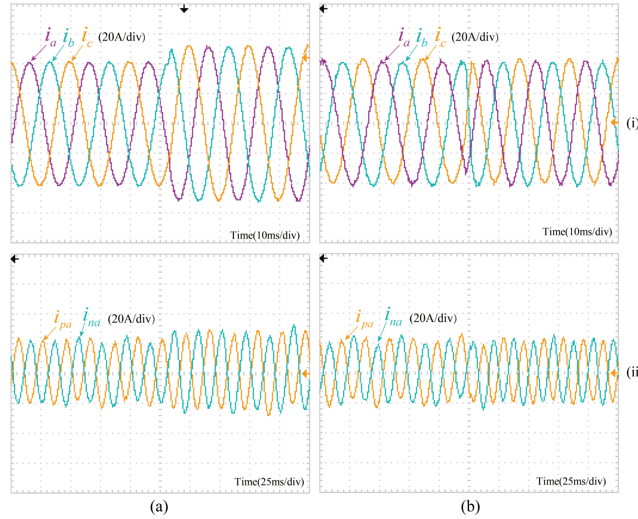

Fig. 10. Transient-state performance of the proposed DD-PCC method: (a) step change in reference current amplitude and (b) step change in reference current frequency.

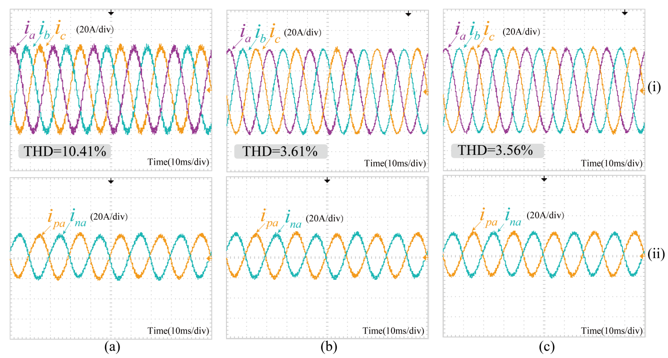

Fig. 11. Robustness performance under parameter mismatch condition. (a) Conventional MPC. (b) ESO-based MPC. (c) Proposed DD-PCC. [(i) three-phase output currents, (ii) upper and lower arm current of phase-a.]

currents and circulating current of each phase can be regulated to their nominal value by only using the input-output data of the controller instead of the explicit information from the mathematical model and actual parameter, resulting in enhanced robustness performance. Another distinctive feature of the proposed methodology is the simplified cost function formula with reduced computational complexity, thereby facilitating the efficient determination of optimal control actions. Finally, simulation and experimental studies were conducted to confirm the feasibility of the proposal.

Due to the simple implementation, the proposed method can be extended to other multilevel converters and different kinds of applications. Depicted that the data-driven predictive current control is capable of rejecting parametric perturbations with an insignificant deterioration of control performance, it is important to emphasize that the proposed method has limitations in addressing some severe external disturbances, such as periodic harmonics and disturbances in the grid side.

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